

python - Day - 15 (Intermediate)

Topic - Local Dev Environment Setup, The Coffee Machine

(*) Local Dev Environment Setup

Instead of coding like in a online editor (like Replit) you'll begin working locally on your computer using a proper code editor/IDE (eg: pycharm, VS Code or IDLE).

This setup prepares you for real-world coding, debugging and file management

Key Setup steps typically include:

- Installing python (if not already)
- Choosing an editor (recommended: pycharm or VS)
- Creating a new python project folder
- Understanding how to run .py files locally
- Using modules, imports and organising files

The focus is on building a coding habit and learning how to actually develop rather than just practicing on a website

(*) Coffee Machine project

Project Goal

Build a Simple text based coffee machine simulation in python that:

- Takes user input for coffee choices
- Tracks ingredient resources
- checks for enough water/milk/coffee
- process coin payments
- prints a report
- Delivers the coffee if successful - or shows errors if not

This consolidates your prior knowledge: variables, input/output, while loops, conditionals, functions and basic logic

Core Concepts you practice

- (1) Control flow (if/elif/else)
- (2) loops (while loop to keep the machine running)
- (3) functions (modularise logic, like payment)
- (4) Dictionaries for menu/ingredient tracking
- (5) Type casting (eg. int()) for coins
- (6) Basic error handling for invalid input
- (7) Coding by planning logic step-by-step before implementation

These fundamentals are essential for large projects later in the course.

Sample Logic Breakdown (Coffee Machine) (*)
while the course video code isn't public
the typical tasks include:

(*) Menu & Resources

```
MENU = {  
    "espresso": {"Ingredients": {"water": 50, "coffee": 18, "milk": 0},  
    "resources": {"water": 300, "milk": 200, "coffee": 100}
```

(*) Taking User Order

```
choice = input("what would you like?: ")
```

(*) Resource checking logic

```
def is_resource_sufficient(order_ingredients):
```

```
    for item in order_ingredients:
```

```
        if order_ingredients[item] > resources[item]:
```

```
            print(f"Sorry there is not enough {item}.")
```

```
            return False
```

```
    return True
```

(*) Process Coins

```
def process_coins():
```

```
    → print("Please insert coins.")
```

```
    total = int(input("how many quarters? ")) * 0.25
```

```
    total += int(input("how many dimes?: ")) * 0.10
```

```
    total += int(input("how many nickels?: ")) * 0.05
```

```
    total += int(input("how many pennies?: ")) * 0.01
```

```
    return total
```

(*) Make Coffee & Update Resources

```
def make_coffee(drink_name, order_ingredients):
    for item in order_ingredients:
        resources[item] -= order_ingredient[item]
    print(f"Here is your {drink_name}!")
```

This structure introduces modular thinking later vital for intermediate and advanced projects like GUIs, games, APIs and web apps.

Summary

Day 15 is a major part of this program

- (1) you switch from beginner Sandboxes to real development
- (2) you build the Coffee Machine project - a real python program with logic and state
- (3) you start thinking like a programmer breaking problems into functions and checking logic step by step.